# Program 7: Build Basic Gradle project

**Step 1:** Create gradle project

gradle init --type java-application

tree

```
harith@harith-VirtualBox:~$ cd devops
harith@harith-VirtualBox:~/devops$ mkdir MyGradleApp
harith@harith-VirtualBox:~/devops$ cd MyGradleApp/
harith@harith-VirtualBox:~/devops/MyGradleApp$ gradle init --type java-application
Starting a Gradle Daemon (subsequent builds will be faster)

BUILD SUCCESSFUL in 15s
2 actionable tasks: 2 executed
harith@harith-VirtualBox:~/devops/MyGradleApp$ tree
.
├── build.gradle
├── gradle
│   └── wrapper
│       ├── gradle-wrapper.jar
│       └── gradle-wrapper.properties
├── gradlew
├── gradlew.bat
├── settings.gradle
└── src
    ├── main
    │   └── java
    │       └── App.java
    └── test
        └── java
            └── AppTest.java

7 directories, 8 files
harith@harith-VirtualBox:~/devops/MyGradleApp$
```

**Step 2:** build the project

gradle build

```
[1]+  Stopped                 gedit build.gradle
harith@harith-VirtualBox:~/devops/MyGradleApp$ gradle build
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.pom
Download https://jcenter.bintray.com/com/google/guava/guava-parent/23.0/guava-parent-23.0.pom
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.po
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_parent/2.0.18/error_prone_parent-2.0.18.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.pom
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-parent/1.14/animal-sniffer-parent-1.14.pom
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.pom
Download https://jcenter.bintray.com/org/codehaus/mojo/mojo-parent/34/mojo-parent-34.pom
Download https://jcenter.bintray.com/org/codehaus/codehaus-parent/4/codehaus-parent-4.pom
Download https://jcenter.bintray.com/com/google/guava/guava/23.0/guava-23.0.jar
Download https://jcenter.bintray.com/com/google/code/findbugs/jsr305/1.3.9/jsr305-1.3.9.jar
Download https://jcenter.bintray.com/com/google/j2objc/j2objc-annotations/1.1/j2objc-annotations-1.1.jar
Download https://jcenter.bintray.com/org/codehaus/mojo/animal-sniffer-annotations/1.14/animal-sniffer-annotations-1.14.jar
Download https://jcenter.bintray.com/com/google/errorprone/error_prone_annotations/2.0.18/error_prone_annotations-2.0.18.ja

BUILD SUCCESSFUL in 20s
7 actionable tasks: 7 executed
```

**Step 3:** edit build.gradle

```
/*
 * This build file was generated by the Gradle 'init' task.
 *
 * This generated file contains a sample Java project to get you started.
 * For more details take a look at the Java Quickstart chapter in the Gradle
 * user guide available at https://docs.gradle.org/4.4.1/userguide/tutorial_java_projects.html
 */

// Apply the java plugin to add support for Java
apply plugin: 'java'

// Apply the application plugin to add support for building an application
apply plugin: 'application'

// In this section you declare where to find the dependencies of your project
repositories {
    // Use jcenter for resolving your dependencies.
    // You can declare any Maven/Ivy/file repository here.
    jcenter()
}

dependencies {
    // This dependency is found on compile classpath of this component and consumers.
    compile 'com.google.guava:guava:23.0'

    // Use JUnit test framework
    testCompile 'junit:junit:4.12'
}

// Define the main class for the application
mainClassName = 'App'
task display{
doLast{
println '6D devops class'
}
}
```

**Step 4:** Run the new task

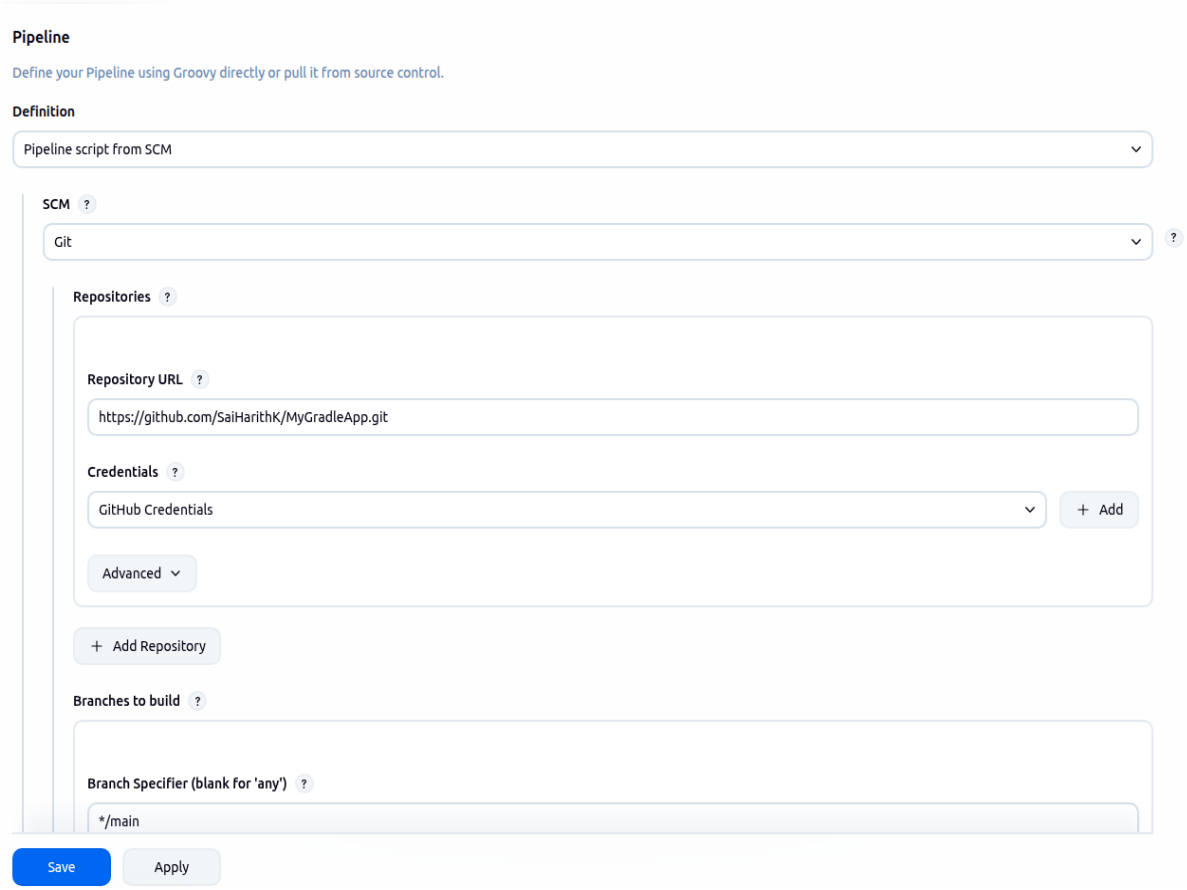
gradle display

```
harith@harith-VirtualBox:~/devops/MyGradleApp$ gradle display

> Task :display
6D devops class


BUILD SUCCESSFUL in 1s
1 actionable task: 1 executed
[2]+  Killed                  gedit build.gradle
harith@harith-VirtualBox:~/devops/MyGradleApp$
```

**Step 6:** Pipeline

-login to jenkins account

-create a new item and configure the following settings

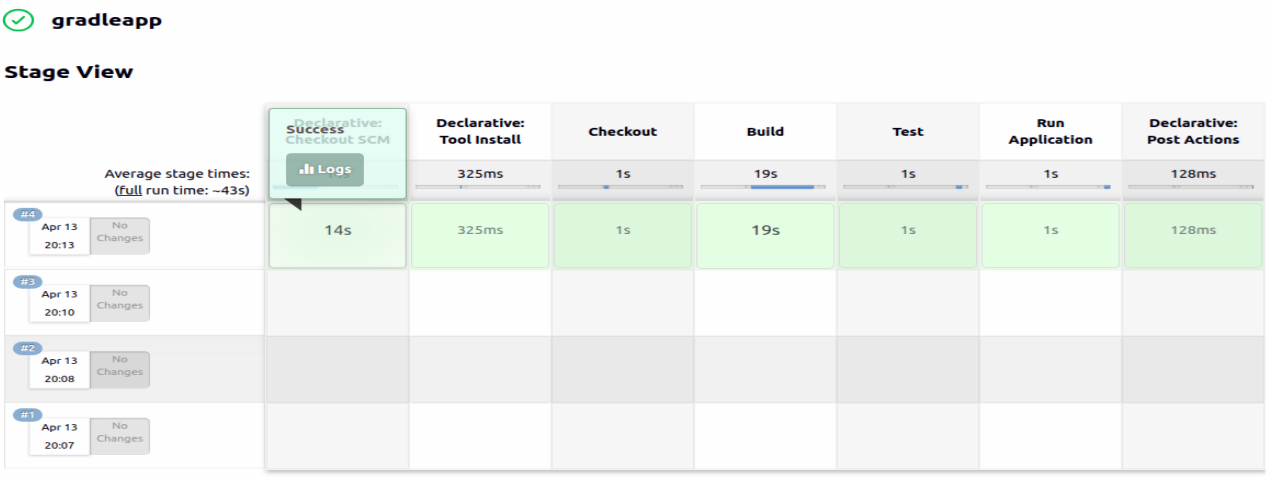



**Step 7:** Edit Jenkins file

-Create a Jenkinsfile and add the following code and push it to github repo

```
pipeline {
    agent any  // Use any available agent
    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
```

```
            steps {
                git branch: 'main', url: 'https://github.com/SaiHarithK/MyGradleApp.git'
            }
        }
        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }
        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
    }
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

**Step 8:** Build

Click on build now and check if its successfull